

EE 474 Introduction to Control Systems**Homework 3**

Due Wednesday September 30, 2026

Reading:

- Franklin, *et al.*: Chapter 2, Sections 2.4 - 2.5
- Lecture notes, Part 2 through end

1 Matlab

In Controls Tutorial for Matlab and Simulink, Read and do the steps of the exercise "Cruise Control / PID" (I recommend that you cut and paste the steps into a .m file, that you can edit and execute.)

Include the plots for each controller (I count 3 distinct plots)

- P-type
- PI-type
- PID-type

Indicate which of the systems has the greatest or least value for these characteristics:

For 10-90 rise time	Fastest / Slowest
Overshoot	Least / Greatest
Steady-state error	Least / Greatest

2 Franklin *et al.* text problems:

A) Text problem 2.11 (2.13 in 8th Ed.)

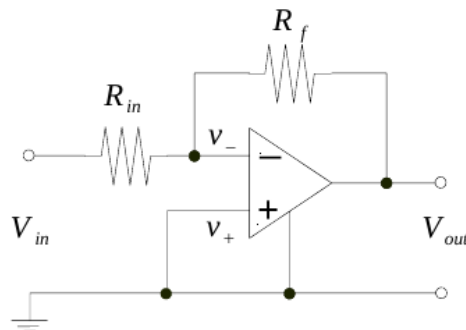
Text problem 2.11 :

A first step toward a realistic model of an op amp is given by the equations below and shown in Fig 2.41.

$$V_{out} = \frac{10^7}{s+1} [V_+ - V_-]$$

$$i_+ = i_- = 0$$

a) Find the transfer function of the simple amplification circuit shown using this model.



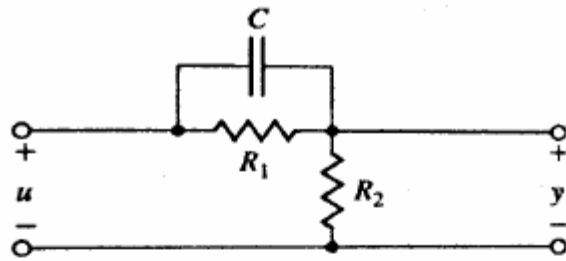
B) Text problem 2.15 (2.17 in 8th Ed.)

Text problem 2.15 :

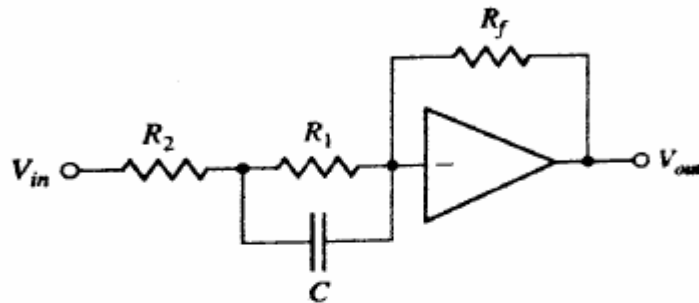
Write the dynamic equations and find the transfer functions for the circuits shown in Fig. 2.51.

- (a) Passive lead circuit
- (b) Active lead circuit
- (c) Active lag circuit
- (d) Passive notch circuit

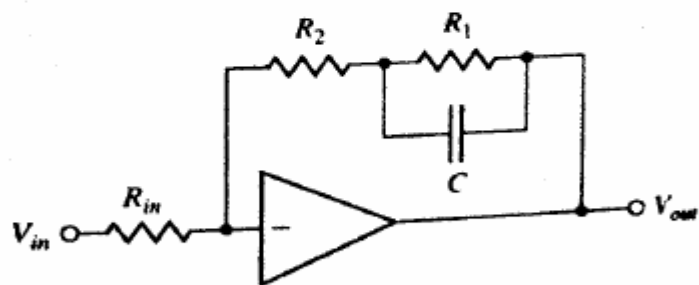
- Do parts a, b, c
- Note: the single input amplifier shown in figure parts (b) and (c) is a regular Op Amp, configured in virtual ground configuration (V^+ input connected to ground). The V^+ input is not shown, which is sometimes done in analog simulation diagrams, and shown with the integrator and summer examples, in the chapter.
- Figure 2.51 is copied on the next page.

Franklin *et al.* Figure 2.51

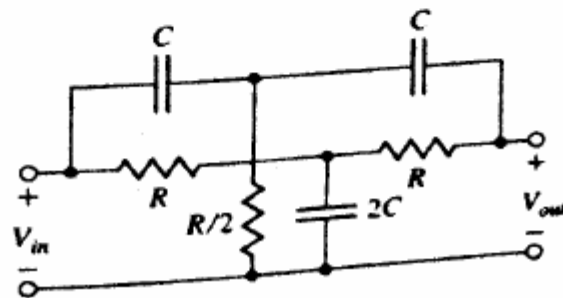
(a)



(b)



(c)



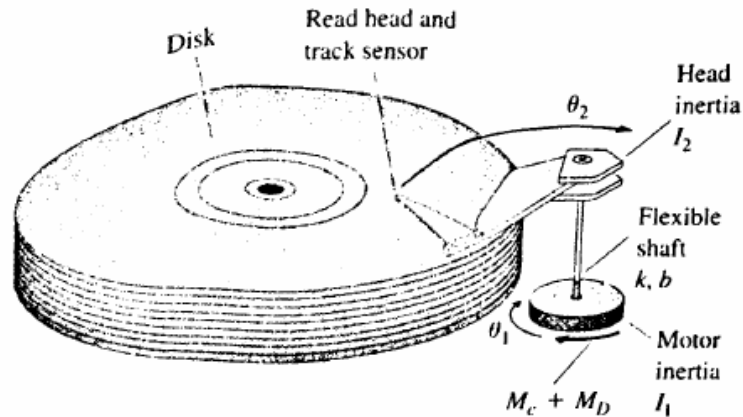
(d)

3 Deriving a transfer function

Derive the transfer function for the Rotational System with Flexibility of Text Example 2.4. (The system is a flexible read/write head in the 7th ed, and Flexible Satellite Attitude Control in the 8th ed. The graphics for the 7th edition are reprinted below.)

The two systems (read/write head and satellite attitude) are analogous, with two rotational inertias plus spring and damper, and lead to the exact same transfer function.

Figure 2.12
Disk read/write head
schematic for modeling



$$I_1 \ddot{\theta}_1 + b(\dot{\theta}_1 - \dot{\theta}_2) + k(\theta_1 - \theta_2) = M_c + M_D, \quad (2.17)$$

$$I_2 \ddot{\theta}_2 + b(\dot{\theta}_2 - \dot{\theta}_1) + k(\theta_2 - \theta_1) = 0. \quad (2.18)$$

Where

M_c is the command torque (moment) [Newton-meters]

M_D is the disturbance torque (for example, friction)

I_1 and I_2 are rotational inertias [Kg-meters²] or, equivalently, $\left[\frac{\text{Newton-meter-seconds}^2}{\text{radian}} \right]$

Derive

$$G_p(s) = \frac{\theta_2(s)}{M_c(s)} \text{ (7th ed.)} \quad , \quad \frac{\theta_2(s)}{T_c(s)} \text{ (8th ed.)}$$

- Show each of steps 1-5 discussed in the lecture notes.

Hint: For systems with multiple inputs, the transfer functions combine by superposition. While deriving $G_p(s)$, the second input signal, if any, such as $M_D(t)$, is considered to be zero.

1. Challenge: can you show the equivalence of [Kg-meters²] and $\left[\frac{\text{Newton-meter-seconds}^2}{\text{radian}} \right]$.